Phishing Awareness Simulation (Educational)

Naviotech Solution - Cyber Security Internship

Group-A SARG - Major Project - Final Report

Project Team

No.	Name	Role
1	Sanika Rajendra Shelar	Team Leader
2	Ayush Singh	Member
3	Roshani Patel	Member
4	Gournag Singh	Member

Abstract

Phishing remains one of the most common attack vectors against individuals and organisations. This project carried out a controlled, consent-based phishing awareness simulation with 20 participants to understand how users behave when they receive a deceptive email, to measure the click rate, and to build awareness. A fake email template annotated with red flags and a safe fake login page (no backend, no storage) were used. Of the 20 participants, 8 (40%) reported the email as phishing, 7 (35%) clicked the link, 4 (20%) entered data on the simulated page, and 1 (5%) ignored the email. Overall, 55% fell for the simulation, which shows a clear need for continued awareness training.

1. Introduction

Phishing is a social engineering technique in which attackers impersonate a trusted party, such as an IT help desk, to trick users into clicking malicious links or revealing credentials. Because it targets human behaviour rather than technical weaknesses, user awareness is one of the most effective defences.

This simulation was conducted ethically, with the informed consent of all 20 participants. Every artefact carried an **EDUCATIONAL SIMULATION - NO DATA COLLECTED** banner.

1.1 Objectives

- Understand user behaviour when faced with a phishing email.
- Measure the click rate of the simulated campaign.
- Create awareness about common phishing red flags.

2. Methodology

The simulation was carried out in five steps.

Step	Activity	Details
1	Consent	Written consent obtained on WhatsApp from all 20 participants.
2	Fake email template	Phishing-style email created with RED FLAG annotations.
3	Safe fake login page	No backend and no storage. Clicking Sign In only displays an awareness message.
4	Distribution	Email sent within a simulated environment.
5	Tracking	Only the click action was counted. No passwords were stored.

3. Simulation Design

3.1 Figure 1: Simulated Phishing Email with Red Flags

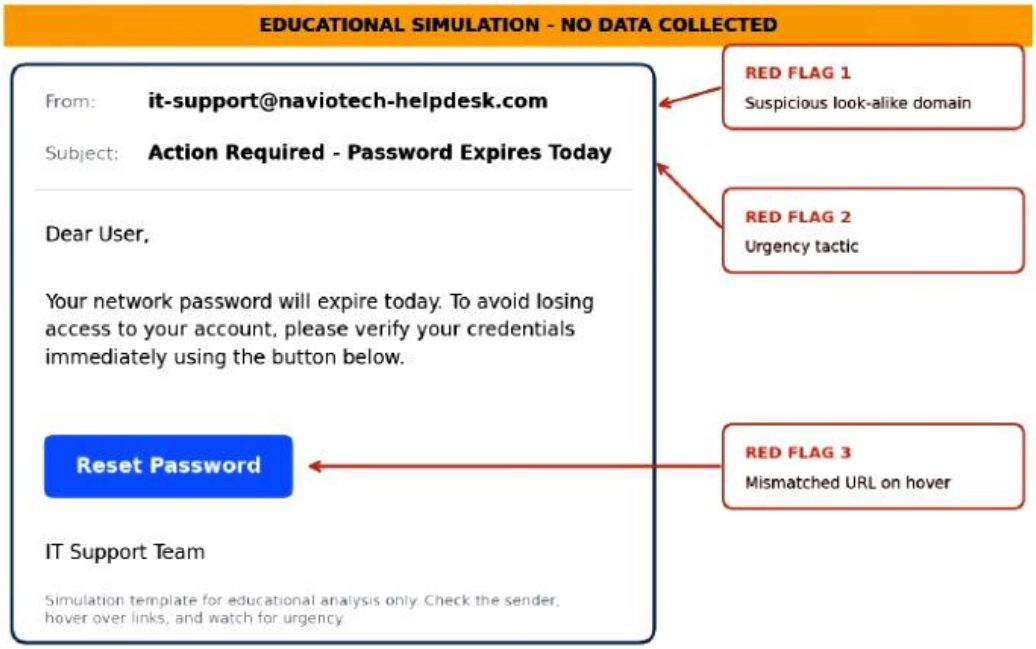


Figure 1: Simulated Phishing Email with Red Flags Annotated - Group-A

The email imitated an IT support message about an expiring password. It contained the following red flags:

Element	Content	Red Flag
Sender	it-support@naviotech-helpdesk.com	Suspicious look-alike domain
Subject	Action Required - Password Expires Today	Urgency tactic
Button	Reset Password	Mismatched URL on hover

The email footer stated that it was a simulation template for educational analysis only, and reminded readers to check the sender, hover over links and watch for urgency.

3.2 Figure 2: Safe Fake Login Page

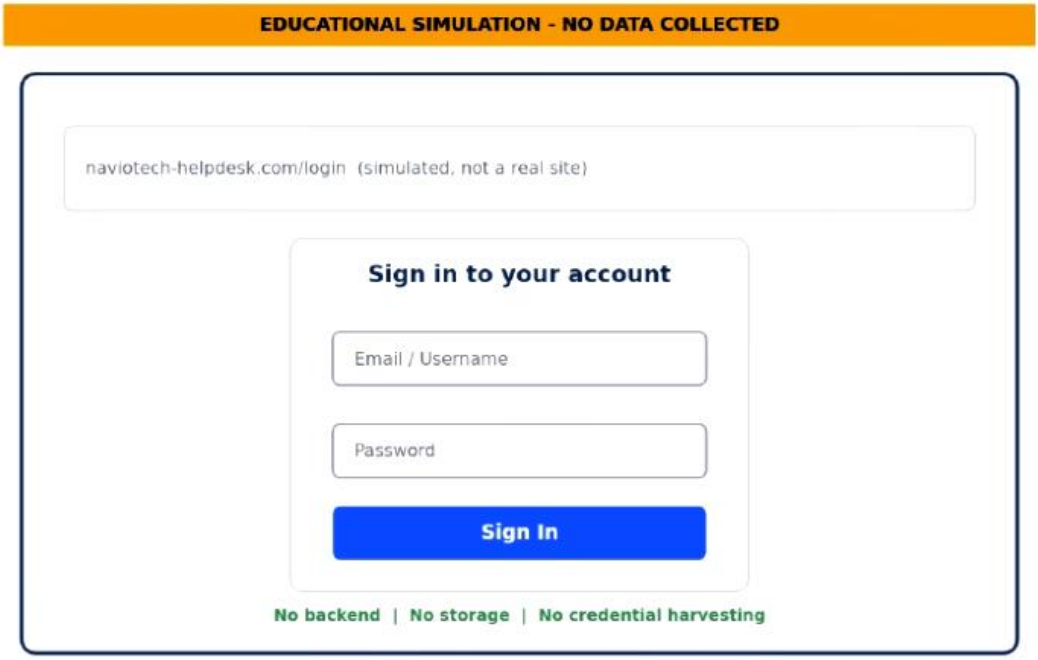


Figure 2: Safe Fake Login Page - Group-A SARG - No Backend, No Storage

The simulated login page had Email/Username and Password fields and a Sign In button. Its safety features were:

- An educational simulation banner at the top of the page.
- No backend, no storage and no credential harvesting.
- On Sign In, the page displays only: *"This was a simulation! You would have been hacked. Always check URL."*

4. Results

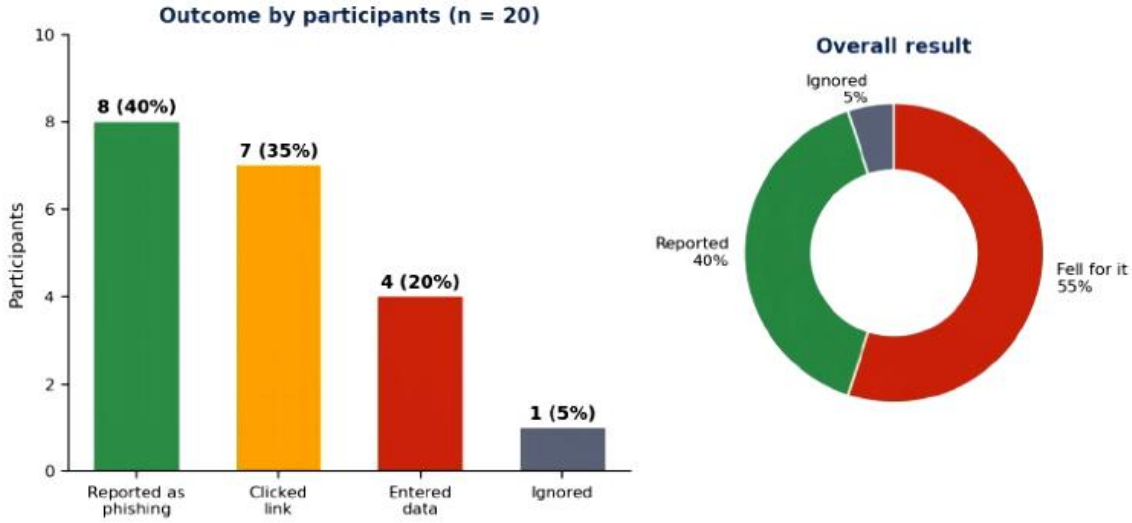


Figure 3: Click Stats - Phishing Simulation Results, 20 participants, with consent

Outcome	Participants	Percentage
Reported as phishing	8	40%
Clicked link	7	35%
Entered data	4	20%
Ignored	1	5%
Total	**20**	**100%**

Overall, 55% of participants fell for the simulation (clicked link plus entered data).

5. Analysis and Lessons Learned

- **35%** clicked the link without checking the URL.
- **20%** went further and entered data on the fake page.
- **40%** correctly reported the email as phishing, which is a good sign.
- Urgency tactics are effective: many users did not verify the sender or the link before acting.
- The reporting culture is promising but needs to improve further.

6. Prevention Measures

6.1 For individuals

- Check the sender's domain carefully.
- Hover over links before clicking.
- Enable two-factor authentication (2FA).
- Never use a link in an email to reset a password. Use the official portal directly.
- Report suspicious emails using the Report Phishing button.

6.2 For organisations

- Implement email authentication controls: DMARC and SPF.
- Run regular phishing awareness training and simulations.
- Encourage and reward reporting of suspicious emails.

7. Ethical Considerations

- All participants gave prior informed consent.
- Every artefact was clearly labelled as an educational simulation.
- The login page had no backend, so no credentials could be captured or stored.
- Only the click action was counted for statistics.

8. Conclusion

The simulation showed that more than half of the participants were susceptible to a basic urgency-based phishing email, while a strong 40% recognised and reported it. Combining regular training, technical controls such as DMARC and SPF, and a healthy reporting culture can significantly reduce phishing risk.

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9. Final Submission Checklist

No.	Item	Status
1	Report PDF with 3 figures	Done
2	PPT with minimum 10 slides (Slide 6: Fig 1 Email, Slide 7: Fig 2 Login, Slide 8: Fig 3 Stats)	
3	GitHub repo *Group-A-Phishing-Awareness-SARG* with PDF, PPT, images and README	
4	Email to info@naviotechsolution.com, subject: *Group-A Final Submission - Sanika Shelar*	

The certificate will be issued after review by the Academic Team.